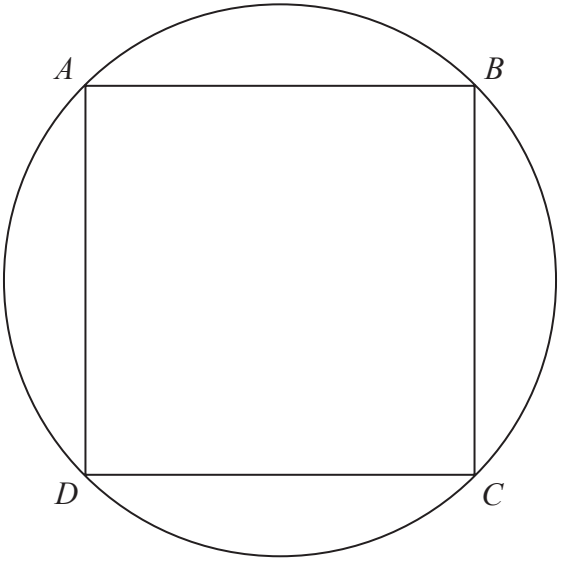
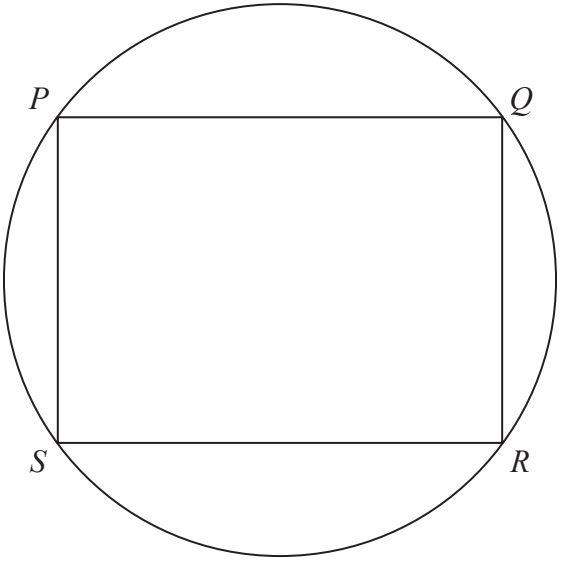
24 *ABCD* is a square with sides 10 cm.
The vertices of the square lie on a circle, $\mathbf{C_1}$



PQRS is a rectangle with sides $(10 + x)$ cm and $(10 - x)$ cm where $0 < x < 10$
The vertices of the rectangle lie on a circle, $\mathbf{C_2}$



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Prove that the area of $\mathbf{C_2}$ is greater than the area of $\mathbf{C_1}$

Each diagonal is a diameter of its circumcircle.

Square: $d^2 = 10^2 + 10^2 = 200$

1 mark
Pythagoras on opposite vertices.

Rectangle: $d^2 = (10 + x)^2 + (10 - x)^2$

$= (100 + 20x + x^2) + (100 - 20x + x^2)$

1 mark
Expand BOTH squares, including both middle terms.

$= 200 + 2x^2$

1 mark
The middle terms cancel.

Circle area $= \pi(d/2)^2 = \pi d^2/4$

Area C2 – area C1 $= \pi(2x^2)/4 = \frac{1}{2}\pi x^2$

Since $0 < x < 10$, $x^2 > 0$, so $\frac{1}{2}\pi x^2 > 0$.

1 mark
The area difference is strictly positive: area C2 > area C1.

(Total for Question 24 is 4 marks)

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